



Modelling Financial Markets

Chapter 10: Market Microstructure



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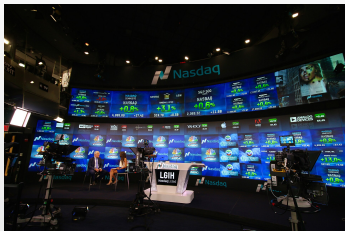
Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AMM: Automated Market Maker
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CARA: Constant Absolute Risk Aversion
- ▣ CEX: Centralised EXchange
- ▣ CFTC: Commodity Futures Trading Commission
- ▣ CI: Confidence Interval
- ▣ DEX: Decentralised EXchange
- ▣ EDGE: Efficient Discrete Generalized Estimator (of the bid-ask spread)
- ▣ EKOP: Easley–Kiefer–O'Hara–Paperman (model of informed trading)
- ▣ ETF: Exchange-Traded Fund
- ▣ FOK: Fill-Or-Kill (order)
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ GMM: Generalized Method of Moments
- ▣ GTC: Good-Till-Cancelled (order)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HFT: High-Frequency Trading
- ▣ IOC: Immediate-Or-Cancel (order)
- ▣ LLM: Large Language Model
- ▣ LOB: Limit Order Book
- ▣ MA: Moving Average
- ▣ MRR: Madhavan–Richardson–Roomans (structural model of trading costs)

Acronyms used in this material (2/2)

- ▣ NYSE: New York Stock Exchange
- ▣ OHLC: Open, High, Low, Close
- ▣ OLS: Ordinary Least Squares
- ▣ PIN: Probability of INformed trading
- ▣ SD: Standard Deviation
- ▣ SE: Standard Error
- ▣ SEC: Securities and Exchange Commission
- ▣ TWAP: Time-Weighted Average Price
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VAR: Vector AutoRegression
- ▣ VECM: Vector Error-Correction Model
- ▣ VIX: CBOE Volatility Index
- ▣ VPIN: Volume-synchronised Probability of INformed trading
- ▣ VWAP: Volume-Weighted Average Price
- ▣ XRP: XRP (the native token of the XRP Ledger)

The Question of This Chapter



Nasdaq MarketSite, Times Square, New York: a studio,
not a trading floor

Photo: Luca Marfè (2015); CC BY 2.0; Wikimedia Commons

- **Question:** what does it cost to trade, and how do prices absorb information, when computers match orders in microseconds?
- The rules of the matching engine (priority, tick size, auctions) shape prices and costs [Gould et al. \(2013\)](#)
- Route
 - ▶ Part 1: order book, spread, illiquidity, intraday patterns
 - ▶ Part 2: information models, price impact and execution, stress, crypto and the BVB, a case study on the speed race
- Textbooks: [Hasbrouck \(2007\)](#), [Bouchaud, Bonart, Donier & Gould \(2018\)](#), [Franke, Härdle & Hafner \(2019\)](#)

Data Map

Series	Content	Period
SPY, 5-minute bars	ETF on the S&P 500; regular session 09:30–16:00 New York time; 1 477 full days	2020–2026
Bitcoin, 5-minute bars	24 hours a day, 7 days a week; 208 596 bars	2024–2026
US stocks, daily	SPY, Apple, Microsoft, Nvidia, JPMorgan, GameStop, Strategy	2016–2026
BVB stocks, daily	Ten blue chips and the BET ETF; values in lei converted to USD at the BNR reference rate	2016–2026
Crypto-assets, daily	Bitcoin, Ethereum, Solana, XRP, Dogecoin, Cardano, Litecoin	2016–2026
VIX, daily	Implied volatility of the S&P 500	2020–2026

- Daily data: open, high, low, close and volume; prices adjusted for dividends and splits; days without trades removed
- Order books in the section *The Limit Order Book* and the bid-ask bounce in the section *The Bid-Ask Spread* are **simulated**, and labelled as such

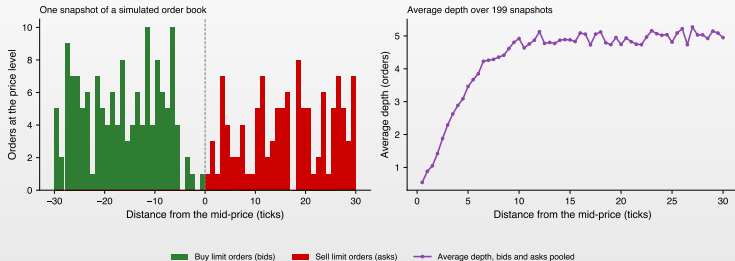
Liquidity and the Limit Order Book: Definitions

- **Liquidity**: trading a quantity quickly at a price close to fair value; no single number captures it
 - ▶ **Tightness** (the spread), **depth** (size needed to move the price), **resiliency** (speed of recovery) Kyle (1985); **immediacy** Harris (2003)
- **Limit order book** (LOB): all standing limit orders, by price and time of arrival Gould et al. (2013)
- Best bid $b_t <$ best ask a_t ; mid-price $m_t = (a_t + b_t)/2$; relative spread $(a_t - b_t)/m_t$ in basis points (1 bp = 0.01%)
 - ▶ **Depth**: quantity waiting at a price level; **tick size**: smallest price step (USD 0.01 for most US stocks above USD 1)

Orders, Priority and Matching

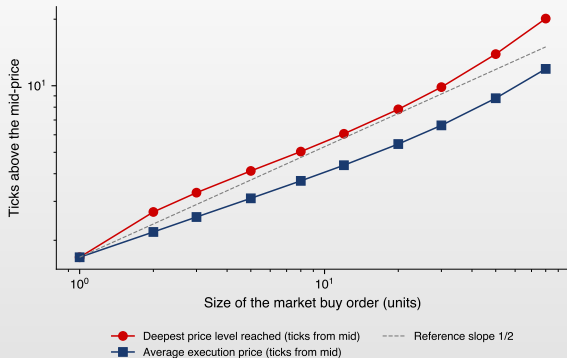
- ▣ **Market order**: immediate execution against the available liquidity, uncertain price; **limit order**: price bounded by the limit, uncertain execution (other order types: [Appendix](#))
- ▣ Resting limit orders supply liquidity; market orders and marketable limit orders consume it
- ▣ **Price-time priority**: the better price first, then the earlier order at the same price (a queue); queue position is valuable (case study: the HFT arms race)
- ▣ **Call auctions** at the open and close: orders are pooled and matched at the single price that maximises the executed quantity
- ▣ A market buy of size Q walks up the ask levels: it pays the half-spread plus the cost of the deeper levels

A Simulated Order Book



- **Simulated:** zero-intelligence order flow [Farmer, Patelli & Zovko \(2005\)](#), limit orders, market orders and cancellations arriving at random (rates: [Appendix](#)); left: one snapshot (bids green, asks red); right: mean depth by distance from the mid
- Mean spread 3.23 ticks, median 3, above 6 ticks 10% of the time; depth 0.8 orders at 1 tick, 4.8 at 10, 5.0 at 25 (199 snapshots)
- Market orders eat the best levels first, so liquidity near the price is scarce [Tóth et al. \(2011\)](#); real books show the same rise away from the mid [Bouchaud, Bonart, Donier & Gould \(2018\)](#)

The Price Impact of a Market Order



- Market buys of 1 to 80 units sent against each simulated snapshot; log-log axes
- Marginal price: a 1-unit order reaches 1.7 ticks above the mid, an 80-unit order 20.1 ticks
- Concave** impact: slope 0.53, close to $1/2$
- Average execution price: slope 0.44; 80 units cost 11.9 ticks on average
- Doubling the order does not double the cost: depth grows with distance from the mid; large orders split over time show a similar law (section *Price Impact and Optimal Execution*)

Why Is There a Spread?

- A liquidity provider buys at the bid and sells at the ask; the spread pays three costs [Huang & Stoll \(1997\)](#)
- **Order processing:** fees, technology, clearing
- **Inventory:** holding an unwanted position exposes the provider to price risk
- **Adverse selection:** some counterparties know more [Bagehot \(1971\)](#)
 - ▶ The provider incurs losses when trading with informed traders and must recover them from uninformed traders (section *Information and Prices*)

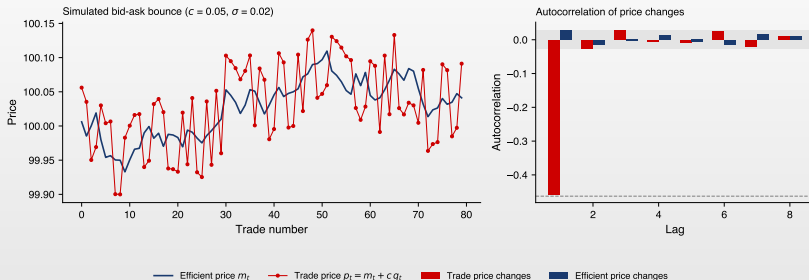
Quoted, Effective and Realised Spread

- ▣ Trade at price p_t , direction $q_t = +1$ (buyer-initiated) or -1 (seller-initiated), mid-price m_t
- ▣ **Quoted:** $S_t^Q = (a_t - b_t)/m_t$; **effective:** $S_t^E = 2q_t(p_t - m_t)/m_t$, twice the signed distance from the mid (round-trip equivalent; the one-way cost is half)
- ▣ **Realised:** $S_t^R = 2q_t(p_t - m_{t+\Delta})/m_t$, the provider's revenue marked to the mid after Δ (e.g. 5 minutes), before fees and inventory costs
 - ▶ Price impact $= S_t^E - S_t^R = 2q_t(m_{t+\Delta} - m_t)/m_t$: the lasting move of the mid after the trade, usually attributed to adverse selection, though inventory and order-flow pressure also contribute
- ▣ Quoted spread: quotes only; effective and realised spread: trades, the prevailing quotes and the trade direction, signed for example by the rule of [Lee & Ready \(1991\)](#)
- ▣ Without such data we **estimate** the spread from prices only: the next slides

The Roll Model

- Efficient price: random walk $m_t = m_{t-1} + \varepsilon_t$, $\varepsilon_t \sim (0, \sigma^2)$ i.i.d. [Roll \(1984\)](#)
 - ▶ Trade price $p_t = m_t + c q_t$, with half-spread c and $q_t = \pm 1$ with probability $1/2$, i.i.d. over time and independent of ε
- Price change: $\Delta p_t = \varepsilon_t + c(q_t - q_{t-1})$
- $\text{Var}(\Delta p_t) = \sigma^2 + 2c^2$, $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2$
- **Roll estimator:** $\hat{s} = 2\hat{c} = 2\sqrt{-\widehat{\text{Cov}}(\Delta p_t, \Delta p_{t-1})}$
 - ▶ The bid-ask bounce creates negative autocorrelation even when the efficient price is a random walk

The Bid-Ask Bounce (Simulated)

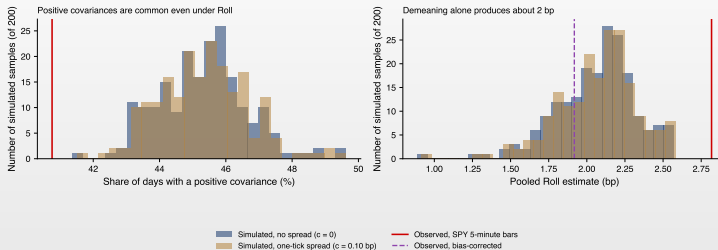


- 5 000 simulated trades with $c = 0.05$, $\sigma = 0.02$; grey band $\pm 1.96/\sqrt{n}$; dashed line: theoretical first-order autocorrelation
- True spread $2c = 0.10$, Roll estimate $\hat{s} = 0.0998$; lag-1 autocorrelation -0.458 against $-c^2/(\sigma^2 + 2c^2) = -0.463$
- No autocorrelation beyond lag 1: the market is efficient, yet trade prices are not a random walk

The Roll Estimator in Finite Samples

- On real data the sample covariance is often **positive**, so $\hat{s}$ does not exist: SPY 5-minute bars, positive on 41% of days
 - ▶ **Sampling error**: the model is true, but T is small and c^2 is small relative to σ^2
 - ▶ **Misspecification**: trends after news, orders split into many trades, stale prices, bars instead of trades
- Demeaning bias [Harris \(1990\)](#): $\mathbb{E}[\hat{\gamma}_1] \approx \gamma_1 - (\gamma_0 + 2\gamma_1)/T$, negative even when $c = 0$; $T \text{Var}(\hat{\gamma}_1) \rightarrow \gamma_0^2 + 3\gamma_1^2$
- Hence $P(\hat{\gamma}_1 > 0) \approx \Phi(\sqrt{T} \gamma_1 / \gamma_0)$, close to 1/2 whenever $c^2 \ll \sigma^2 / \sqrt{T}$
- Delta-method standard error and the daily-data case: [Appendix](#); seminar A1–A2 derive them step by step

SPY: Sampling Error or Misspecification?



- Per day 77 five-minute returns; a one-tick spread means a half-spread $c = 0.10$ bp against a bar standard deviation of 9.2 bp; 200 simulated samples of the same size under the Roll model
- One-tick spread: $\hat{\gamma}_1 > 0$ on 45.3% of days [43.0, 48.1]; observed 40.8%, below the band: more negative autocorrelation than a one-tick bounce
- No spread ($c = 0$): the Roll estimator from the autocovariance averaged over all days is still 2.06 bp [1.50, 2.53], so most of the observed 2.82 bp is demeaning bias; bias-corrected 1.92 bp points to transient price pressure, not the bounce

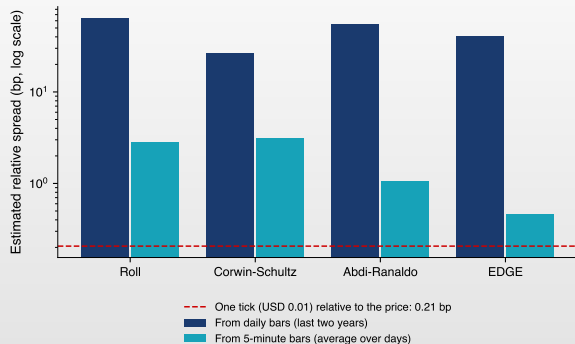
Spreads from Daily Ranges: Corwin–Schultz

- Idea: the daily high is usually a buy at the ask, the low a sell at the bid; the high-low range = volatility + spread [Corwin & Schultz \(2012\)](#)
- The volatility part grows with the length of the interval, the spread part does not: compare one day with two days
- $\beta = \sum_{j=0}^1 \left[\ln \frac{H_{t+j}}{L_{t+j}} \right]^2$, $\gamma = \left[\ln \frac{H_{t,t+1}}{L_{t,t+1}} \right]^2$, $\alpha = \frac{\sqrt{2\beta} - \sqrt{\beta}}{3 - 2\sqrt{2}} - \sqrt{\frac{\gamma}{3 - 2\sqrt{2}}}$
- $\hat{S} = \frac{2(e^\alpha - 1)}{1 + e^\alpha}$, with $H_{t,t+1}$, $L_{t,t+1}$ the high and low over the two days; negative two-day estimates are set to 0, then averaged
- For daily data the paper adjusts for overnight moves ([Appendix](#))

Abdi-Ranaldo and EDGE

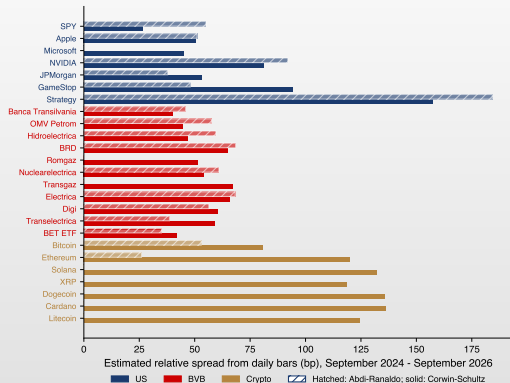
- **Abdi-Ranaldo** [Abdi & Ranaldo \(2017\)](#): the close c_t (log) sits at the bid or ask; the mid-range $\eta_t = (\ln H_t + \ln L_t)/2$ proxies the mid-price
 - ▶ $\hat{S}^2 = 4 \mathbb{E} [(c_t - \eta_t)(c_t - \eta_{t+1})]$
- **EDGE** [Ardia, Guidotti & Kroencke \(2024\)](#): uses open, high, low and close; two moment conditions, each a product of log-returns between the open, the close and η_t , give unbiased estimates $\hat{e}_1, \hat{e}_2$ of S^2 with variances v_1, v_2
 - ▶ Minimum-variance combination (diagonal weighting): $\hat{S}^2 = (v_2 \hat{e}_1 + v_1 \hat{e}_2) / (v_1 + v_2)$
- Common to both: only OHLC bars are needed; a negative mean moment gives $\hat{S} = 0$; $\sqrt{\hat{S}^2}$ is not exactly unbiased for S
- For crypto-assets, high-low estimators track the time variation of high-frequency liquidity; the Amihud ratio is better for levels [Brauneis, Mestel, Riordan & Theissen \(2021\)](#)

Spread Estimators at Two Frequencies: SPY



- Daily bars Sep 2024 – Sep 2026; 5-minute bars, 1 477 days (2020–2026)
- One tick (USD 0.01) is 0.21 bp: a floor for the **quoted** spread, not for the effective one
- Daily: Roll 64.3, Corwin–Schultz 26.5, Abdi–Ranaldo 55.1, EDGE 40.1 bp: two orders of magnitude above a tick, volatility dominates [Jahan-Parvar & Zikes \(2023\)](#)
- 5-minute: Roll 2.82, Corwin–Schultz 3.11, Abdi–Ranaldo 1.06 bp; EDGE 0.46 bp [0.42, 0.49], the closest to a tick
- The windows differ; on the same two years the 5-minute Corwin–Schultz is 2.83 bp (seminar B2); a tick is a plausibility check, not a validation

Spread Estimates Across 25 Assets



- Daily bars, Sep 2024 – Sep 2026; BVB stocks in lei, prices adjusted for dividends and splits
- Median Corwin–Schultz: US 53, BVB 54, crypto 124 bp
- BVB blue chips look as liquid as US large caps: implausible, since they trade a thousand times less value (next section)
- The ranking follows volatility; Abdi–Rinaldo is 0 for 8 of the 25 assets (negative sample mean)
- Levels need an independent benchmark: the effective spread measured from trades [Jahan-Parvar & Zikes \(2023\)](#)

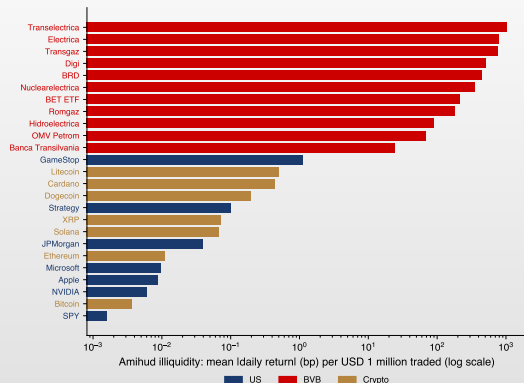
The Amihud Illiquidity Ratio

- Daily price impact per unit of money traded [Amihud \(2002\)](#)

►
$$\text{ILLIQ}_i = \frac{1}{D_i} \sum_{d=1}^{D_i} \frac{|r_{i,d}|}{\text{DVOL}_{i,d}}, \text{ with } r_{i,d} \text{ the daily return and } \text{DVOL}_{i,d} \text{ the traded value}$$

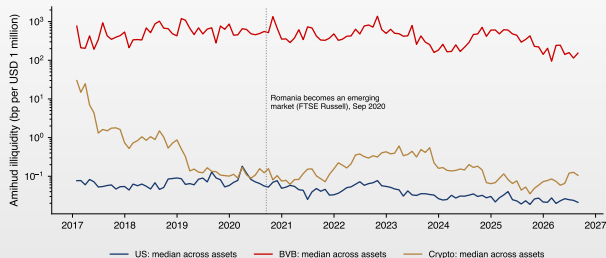
- Units here: basis points of price move per USD 1 million traded
- Needs only daily returns and volumes: available for every market and every period
- A low-frequency proxy of the Kyle price-impact coefficient (section *Information and Prices*) [Hasbrouck \(2009\)](#)
- Traded value of BVB stocks: value in lei divided by the BNR reference rate of the day; crypto: traded value in USD summed across exchanges

Amihud Illiquidity Across 25 Assets



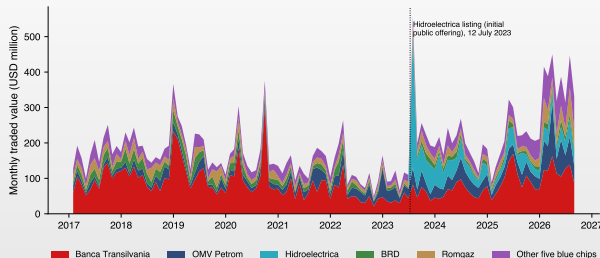
- Sep 2024 – Sep 2026, log scale; mean over about 500 trading days (730 for crypto)
- SPY 0.0016 bp per USD 1 million (median value USD 40 124 million a day); Bitcoin 0.0037 bp
- Banca Transilvania, the most liquid BVB stock: 24.7 bp, about 15 541 times SPY; Transelectrica 1024.0; the BET ETF 212.9
- Three clusters: US large caps and top crypto, smaller crypto and GameStop, the BVB; unlike daily spread proxies, Amihud separates markets by orders of magnitude
- A proxy, not a cost you could actually trade at

Illiquidity over Time: Three Markets



- Monthly Amihud ratio, median across each group (without SPY and the BET ETF), log scale
- Crypto: from 7.6 bp per USD 1 million (2017) to 0.089 (2026); US: from 0.062 to 0.023
- BVB: from 435 to 169, still about 7 212 times the US median; no break at the move to emerging-market status (September 2020) [BVB \(history\)](#)

Traded Value of BVB Blue Chips

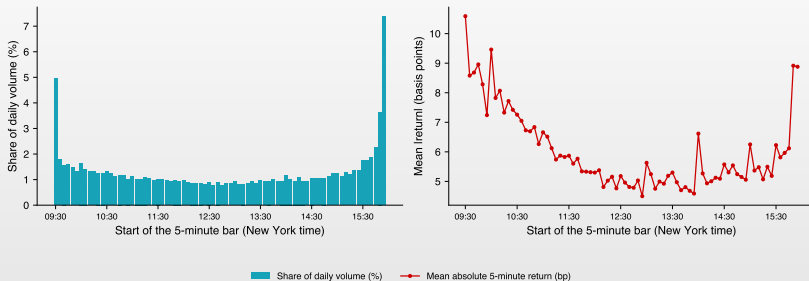


- Monthly traded value of ten blue chips, in USD at the BNR reference rate: 2 098 million in 2017, 1 494 million in 2022, 2 520 million in 2025
- Concentration: Banca Transilvania 44% of the 2025 value, Hidroelectrica (listed July 2023) 13%
- One day of SPY exceeds a year of the ten blue chips: a position of a few million USD takes days to build or unwind

Is Liquidity Priced?

- **Level:** investors demand a higher return for assets that are costly to trade [Amihud & Mendelson \(1986\)](#)
 - ▶ Expected return increases with the spread, concavely: long-horizon investors hold the illiquid assets
- **Time series:** expected market illiquidity raises future returns; an unexpected rise lowers current prices [Amihud \(2002\)](#)
- **Liquidity risk:** stocks that fall when market liquidity dries up earn a premium [Pástor & Stambaugh \(2003\)](#), [Acharya & Pedersen \(2005\)](#)
- Evidence here: monthly market returns fall when illiquidity rises unexpectedly on the BVB, in the US and in crypto ($t = -4.0, -3.8, -4.1$; seminar C1)

SPY Within the Trading Day

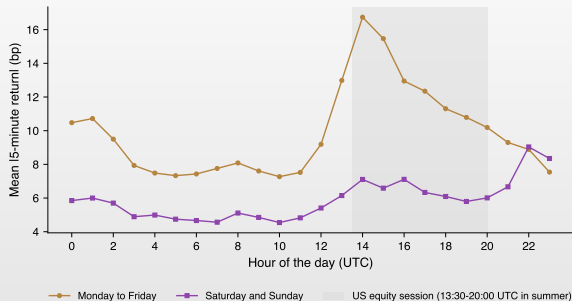


- 78 five-minute bars a day, 1 477 full days (Oct 2020 – Sep 2026); the first bar starts at the opening price
- Volume: first bar 5.0% of the day, a midday bar 0.9%, last 30 minutes 18.6% [18.3, 19.0] against 7.7% if even
- Mean $|r|$: 10.6 bp in the first bar, 5.1 bp at midday, ratio 2.06 [1.98, 2.15] (95% CI resampling whole days, seminar B1); the spikes at 10:00 and 14:00 coincide with scheduled macroeconomic data releases

Why a U-Shape?

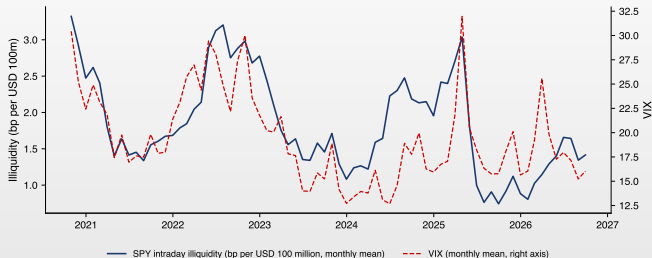
- Documented for NYSE stocks by [Wood, McInish & Ord \(1985\)](#): returns and volume are highest at the open and the close
- **Open**: overnight news is absorbed; uncertainty is highest
 - ▶ Informed traders act early, before the information spreads
- **Clustering**: uninformed traders who can choose the time trade together, and liquidity attracts liquidity [Admati & Pfleiderer \(1988\)](#)
- **Close**: index funds and ETFs trade at the closing price, which is fixed by the closing auction
- Practical rule: an execution algorithm follows the volume curve to reduce impact (section *Price Impact and Optimal Execution*)

Bitcoin Around the Clock



- Mean absolute 5-minute return by hour (UTC), Sep 2024 – Sep 2026, 208 596 bars
- Weekdays: peak 16.7 bp at 14:00 UTC (US open, macro releases), low 7.3 bp at 10:00 UTC
- Weekend 5.9 bp against 9.9 bp on weekdays: 1.69 times calmer (confidence intervals: seminar B7)
- A 24/7 market still follows the clock of banks, ETFs and traditional investors

Liquidity and Volatility: SPY and the VIX



- Daily intraday illiquidity: mean over the 78 bars of $|r|$ / traded value, bp per USD 100 million; monthly means shown
- Correlation with the VIX: 0.58 daily, 0.69 for monthly means; elasticity of log illiquidity on log VIX 0.95 (HAC SE 0.12; fitted relation in seminar B4)
- April 2025: 3.04 against a sample mean of 1.83: liquidity dries up when investors most need it, so liquidity risk is correlated with market risk [Acharya & Pedersen \(2005\)](#)

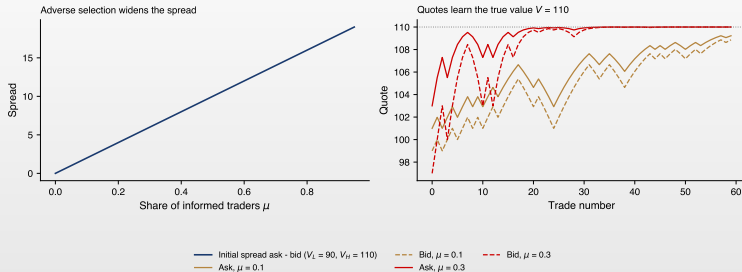
The Glosten–Milgrom Model

- Value $V \in \{V_L, V_H\}$, $P(V = V_H) = \theta$; each period one trader arrives [Glosten & Milgrom \(1985\)](#)
 - ▶ With probability μ informed (buys if $V = V_H$, sells if $V = V_L$), otherwise uninformed, trading for liquidity needs (buys or sells with probability $1/2$) [Bagehot \(1971\)](#)
- Competitive, risk-neutral market maker: zero expected profit on each trade
- $a = \mathbb{E}[V \mid \text{buy}]$, $b = \mathbb{E}[V \mid \text{sell}]$
- $P(\text{buy} \mid V_H) = \mu + \frac{1-\mu}{2}$, $P(\text{buy} \mid V_L) = \frac{1-\mu}{2}$
- After each trade, Bayes' rule updates θ : a buy raises it, a sell lowers it

Glosten–Milgrom: the Spread

- $P(V_H \mid \text{buy}) = \frac{\left(\mu + \frac{1-\mu}{2}\right) \theta}{\left(\mu + \frac{1-\mu}{2}\right) \theta + \frac{1-\mu}{2}(1-\theta)}, \quad a = V_L + (V_H - V_L) P(V_H \mid \text{buy})$
- At $\theta = 1/2$: $a - b = \mu(V_H - V_L)$
 - ▶ The spread grows linearly with the share of informed traders
 - ▶ $V_L = 90$, $V_H = 110$: spread 2 for $\mu = 0.1$, 6 for $\mu = 0.3$
- No inventory, no fees: the spread exists purely because of information
- Trade prices form a martingale: the market is efficient in the semi-strong sense ([Chapter 2](#))

Glosten–Milgrom in Action



- Left: initial spread as a function of μ ; right: one simulated sequence of 60 trades with true value $V = 110$ (same random draws for both μ)
- On these individual paths the bid exceeds 108 after 7 trades for $\mu = 0.3$, only after 48 for $\mu = 0.1$
- More informed traders: wider spreads, but faster price discovery, a trade-off between liquidity and informational efficiency; empirical counterpart: PIN [Easley, Kiefer, O'Hara & Paperman \(1996\)](#)

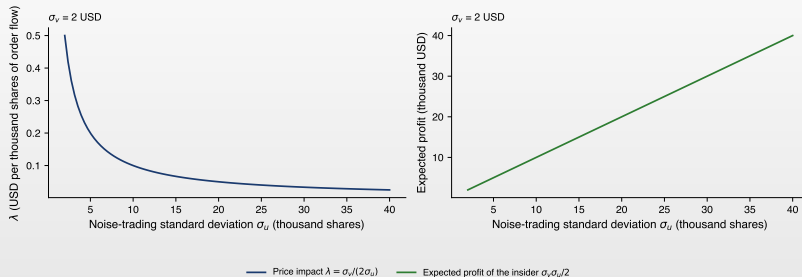
The Kyle Model

- One insider knows $v \sim N(p_0, \sigma_v^2)$ and submits a market order x [Kyle \(1985\)](#)
 - ▶ Noise traders submit $u \sim N(0, \sigma_u^2)$, independent of v
 - ▶ The market maker sees only the total order flow $y = x + u$ and sets $p = \mathbb{E}[v \mid y]$
- The insider chooses x to maximise $\mathbb{E}[(v - p)x]$, knowing that his own order moves the price
- Linear equilibrium: $x = \beta(v - p_0)$, $p = p_0 + \lambda y$

Kyle: Derivation and Equilibrium

- **Insider:** given $p = p_0 + \lambda(x + u)$, maximise $\mathbb{E}[(v - p)x \mid v] = (v - p_0)x - \lambda x^2$
 - ▶ First-order condition: $x = (v - p_0)/(2\lambda)$, so $\beta = 1/(2\lambda)$: he trades less because his order moves the price
- **Market maker:** (v, y) is jointly Normal, so $\mathbb{E}[v \mid y]$ is the linear projection
 - ▶ $\lambda = \text{Cov}(v, y)/\text{Var}(y) = \beta\sigma_v^2/(\beta^2\sigma_v^2 + \sigma_u^2)$ (the population OLS slope of v on y)
- Substituting $\beta = 1/(2\lambda)$ gives $\beta^2\sigma_v^2 = \sigma_u^2$
- **Equilibrium**
 - ▶ $\beta = \sigma_u/\sigma_v$, $\lambda = \sigma_v/(2\sigma_u)$, $\mathbb{E}[\text{insider profit}] = \sigma_v\sigma_u/2$
 - ▶ $\text{Var}(x) = \sigma_u^2$: the insider hides behind noise traders; $\text{Var}(v \mid y) = \sigma_v^2 - \lambda^2\text{Var}(y) = \sigma_v^2/2$: the price reveals half of the private information

Kyle's Lambda and the Insider's Profit



- $\sigma_v = 2$ USD; noise trading σ_u from 2000 to 40 000 shares
- λ : price move per share of order flow (chart: USD per thousand shares), a measure of informativeness; $1/\lambda$ is **market depth**
- More noise trading: a deeper market, larger insider trades and profit, while the posterior variance stays $\sigma_v^2/2$

From Models to Measurement

Measure	What it identifies	Data needed
Kyle λ	price change per unit of signed order flow; the Amihud ratio is its daily proxy Hasbrouck (2009)	signed flow, or daily bars
MRR	permanent (adverse-selection) and transitory parts of the spread Madhavan, Richardson & Roomans (1997)	signed trades and quotes
Hasbrouck VAR	long-run quote response to a trade innovation Hasbrouck (1991)	signed trades and quotes
PIN	$\alpha\mu/(\alpha\mu + \varepsilon_b + \varepsilon_s)$, the expected share of informed trades Easley, Kiefer, O'Hara & Paperman (1996)	daily buy and sell counts
Order-flow imbalance	explains most short-horizon price changes Cont, Kukanov & Stoikov (2014)	order-book events

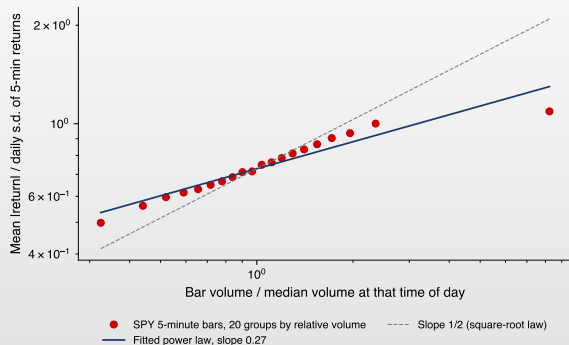
- VPIN, order imbalance in volume time [Easley, López de Prado & O'Hara \(2012\)](#), is contested: correctly computed it did not predict the Flash Crash and largely tracks volume and volatility [Andersen & Bondarenko \(2014\)](#)
- The public bars used here do not sign trades, so MRR, the VAR and PIN are not estimated here; equations and likelihood:

► [Appendix](#)

Metaorders and the Square-Root Law

- Signs of market orders are autocorrelated for hundreds of trades: large **metaorders** are split into many small trades over hours or days [Bouchaud, Gefen, Potters & Wyart \(2004\)](#) (propagator and Hawkes models: 
▶ Appendix)
- Empirical law for the peak impact (from the first to the last trade of the metaorder) [Tóth et al. \(2011\)](#), [Bouchaud, Bonart, Donier & Gould \(2018\)](#)
 - ▶ $\Delta P \approx Y \sigma_{\text{day}} \sqrt{Q/V_{\text{day}}}$, with Q the metaorder size, V_{day} the daily volume, Y of order 1
- Concave: trading 1% of daily volume moves the price by about 0.1σ , not 0.01σ : liquidity near the price is scarce and refills slowly (section *The Limit Order Book*)
- Cost: if impact builds as $I(q) = Y \sigma \sqrt{q/V}$ during execution, the average price paid is $\frac{2}{3} I(Q)$ (before spread and fees)
- Testing it needs the traders' own order records; public bars allow only an indirect look (next slide)

Volume and Price Moves in SPY Bars



- 115 052 five-minute bars in 20 equal-size groups by relative volume (bar volume / median volume at that time of day)
- Mean $|r|$, scaled by the day's volatility, rises with slope 0.27 on log-log axes: concave, but flatter than 1/2
- An association between bars, not a metaorder law: news drives both volume and $|r|$ (simultaneity), and opposite trades in a bar net out
- Seminar B6 tries an instrumental variable and questions its exclusion restriction; for execution costs calibrate the metaorder law on your own trades

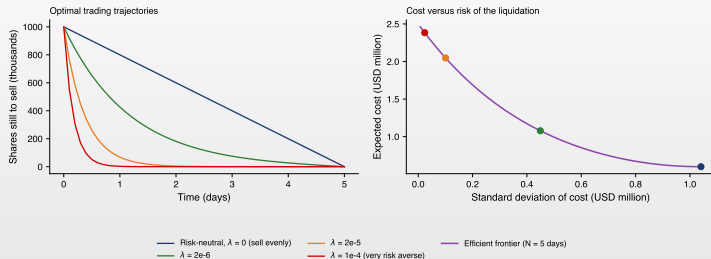
Optimal Execution: the Trade-Off

- Sell X shares within T days, in N steps of length $\tau = T/N$; holdings x_k with $x_0 = X$, $x_N = 0$; trades $n_k = x_{k-1} - x_k$
Almgren & Chriss (2001)
- Price dynamics: $S_k = S_{k-1} + \sigma\sqrt{\tau}\xi_k - \gamma n_k$; execution price $\tilde{S}_k = S_{k-1} - \eta n_k/\tau$; ξ_k i.i.d., mean 0, variance 1
 - γ : permanent impact; η : temporary impact, paid only by the trade itself
- **Trading fast**: high temporary impact, low price risk; **trading slowly**: the opposite
- Objective: minimise $\mathbb{E}[C] + \lambda \text{Var}[C]$ over deterministic schedules, with C the implementation cost and $\lambda \geq 0$ the risk aversion

Almgren–Chriss: the Optimal Trajectory

- Continuous time: minimise $\int_0^T (\eta \dot{x}^2 + \lambda \sigma^2 x^2) dt$ with $x(0) = X$, $x(T) = 0$; solution $x(t) = X \sinh(\kappa(T - t)) / \sinh(\kappa T)$, $\kappa^2 = \lambda \sigma^2 / \eta$ (derivation: [Appendix](#), seminar A7)
- Discrete time, x_k the holdings after trade k ($t_k = k\tau$), $n_k = x_{k-1} - x_k$:
 - ▶ $x_k = X \frac{\sinh(\kappa(T - t_k))}{\sinh(\kappa T)}$, $\frac{2}{\tau^2} (\cosh \kappa \tau - 1) = \frac{\lambda \sigma^2}{\tilde{\eta}}$, $\tilde{\eta} = \eta - \frac{\gamma \tau}{2} > 0$
 - ▶ $\mathbb{E}[C] = \frac{1}{2} \gamma X^2 + \frac{\tilde{\eta}}{\tau} \sum_{k=1}^N n_k^2$, $\text{Var}[C] = \sigma^2 \tau \sum_{k=1}^N x_k^2$
- $\lambda = 0$: $\kappa = 0$ and holdings fall linearly (trade evenly); large λ : most of the order is traded at the start, cutting the exposure to price risk
- $1/\kappa$ is the characteristic time scale of the liquidation

Optimal Trajectories and the Efficient Frontier



- Numerical example of [Almgren & Chriss \(2001\)](#): $X = 10^6$ shares, $S_0 = 50$ USD, $T = 5$ days, $\sigma = 0.95$ USD per day^{1/2}, $\eta = 2.5 \cdot 10^{-6}$, $\gamma = 2.5 \cdot 10^{-7}$
- With $\lambda = 2 \cdot 10^{-6}$ the first day sells 57% of the position; the SD of cost falls from 1.04 (even selling) to 0.45 million USD for 0.48 million USD more expected cost
- The frontier is convex: the first units of risk reduction are cheap, the last ones expensive (full table: [Appendix](#))

Execution Benchmarks

- ▣ A **benchmark** measures execution quality; an **execution rule** decides what to trade when
- ▣ **Implementation shortfall**: paper portfolio at the decision price minus the executed portfolio [Perold \(1988\)](#)
- ▣ VWAP: the volume-weighted average price over the period; TWAP: the time-weighted one, matched by the $\lambda = 0$ schedule
- ▣ A participation rule (a fixed share of market volume) ignores price: the rule at the heart of the Flash Crash (next section)

High-Frequency Trading: the New Market Makers

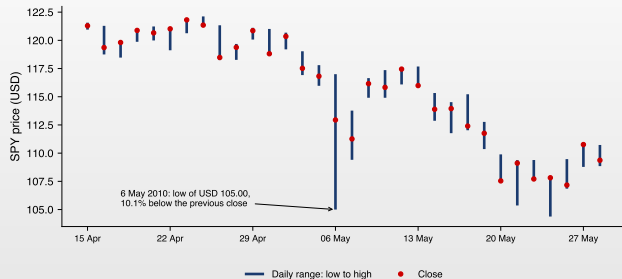


A microwave relay tower (Mojave, California); similar line-of-sight links carry trading signals between Chicago and New Jersey

Photo: [Tony Webster \(2016\)](#); CC BY-SA 2.0; [Wikimedia Commons](#)

- HFT: algorithms that trade in micro- to milliseconds and end the day flat; speed from co-location (servers placed next to the exchange's matching engine) and microwave links
- Low latency speeds up price discovery and narrows spreads [Hasbrouck & Saar \(2013\)](#)
- One HFT firm made markets on Chi-X and Euronext, mostly passively: it earned the spread and lost on positions [Menkveld \(2013\)](#)
 - ▶ Order flow and latency become “information” [O'Hara \(2015\)](#); market makers as intermediaries: [Pele \(2025\)](#)
- Inventory skews quotes: a dealer who is long lowers both quotes to sell [Avellaneda & Stoikov \(2008\)](#) ([▶ Appendix](#))
- HFT liquidity is plentiful in calm markets and fragile under stress

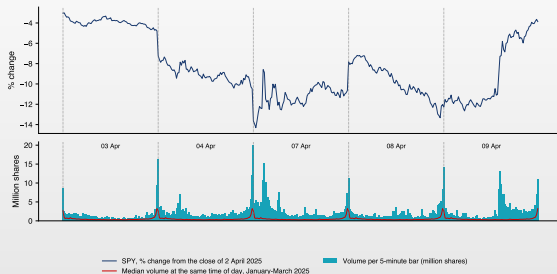
The Flash Crash of 6 May 2010



- **Daily SPY range and close, 15 Apr – 28 May 2010 (unadjusted):** the bars cannot show the intraday collapse and recovery
- 2:32 p.m.: a mutual fund complex starts selling 75 000 E-mini S&P 500 futures contracts at 9% of volume, regardless of price; 2:45:28 p.m.: a 5-second pause lets buyers return [CFTC & SEC \(2010\)](#)
- SPY low USD 105.00, 10.1% below the previous close of 116.82; close only 3.3% lower; volume 3.0 times the recent median
- HFT did not trigger the crash but amplified it [Kirilenko, Kyle, Samadi & Tuzun \(2017\)](#); responses: single-stock circuit breakers, rules for erroneous trades (details: [Appendix](#))

 **MFM_ch10_stress**

A Recent Stress Episode: April 2025



- SPY 5-minute bars, 3–9 April 2025; bottom: bar volume against the Jan–Mar 2025 median at the same time of day
- 7 April: range 8.6%, one bar +4.28% and one ten minutes later −2.92%, realised volatility 6.7% (median 0.61%); 9 April: +10.0% open to close; volume 4.6 times usual
- Bars document sharp moves and heavy trading; they cannot establish book depth, price continuity or the absence of halts

Crypto Venues and Arbitrage

- 24/7 trading with no auctions; the same asset trades on many CEX order books at once, and on DEX pools priced by a formula (AMM, [Chapter 16](#))
 - ▶ The same Bitcoin can trade at different prices on different exchanges [Makarov & Schoar \(2020\)](#)
- Price gaps are much larger **across countries** than within a country
 - ▶ Capital controls limit the movement of arbitrage capital
- A common component of signed volume across exchanges explains 80% of Bitcoin returns: price discovery is shared across venues [Makarov & Schoar \(2020\)](#)
- Crypto liquidity can be forecast from market data [Agakishiev, Härdle & Pele \(2025\)](#)

Price Discovery across Venues: Information Shares

- Two prices of one asset, $y_t = (p_{1t}, p_{2t})'$, are $I(1)$ and cointegrated with $\beta = (1, -1)'$; VECM (vector error-correction model):
 - ▶ $\Delta y_t = \alpha \beta' y_{t-1} + \sum_{i=1}^p \Gamma_i \Delta y_{t-i} + e_t$, $\text{Var}(e_t) = \Omega$ [Johansen \(1991\)](#)
 - ▶ Common efficient price: the stochastic trend with loadings $\psi' \propto \alpha'_\perp$, where $\alpha'_\perp \alpha = 0$
- **Hasbrouck information share**: venue j 's share of the variance of the common-trend innovation, $([\psi' F]_j)^2 / \psi' \Omega \psi$, F the Cholesky factor of Ω [Hasbrouck \(1995\)](#)
 - ▶ Depends on the ordering when Ω is not diagonal: report the bounds over both orderings
- **Gonzalo–Granger component share**: $\alpha_\perp$ normalised to sum 1; in $[0, 1]$ only if the two adjustment coefficients have opposite signs [Gonzalo & Granger \(1995\)](#)
- A price that does **not** adjust ($\alpha_j = 0$, weak exogeneity) leads price discovery; the inference is a Wald test of $\alpha_j = 0$, and non-rejection does not prove $\alpha_j = 0$

Price Discovery on the BVB: the BET ETF and the BET-TR Index

- ▣ Daily log closes of the Patria-TV BET ETF (it reinvests dividends) and the BET-TR index, 495 common trading days, September 2024 – September 2026; the basis (log ETF minus log index) has standard deviation 0.84%; BET-TR is computed from its constituents' prices, not traded on a venue
- ▣ Johansen trace test: 49.6 for $r = 0$ (5% critical value 15.49), 0.10 for $r \leq 1$ (3.84): one cointegrating vector; VECM with 1 lag (BIC)
- ▣ Adjustment: ETF $\alpha = -0.12$ ($t = -2.11$); index $\alpha = -0.03$ ($t = -0.63$)
 - ▶ The index adjustment is not distinguishable from zero, the ETF's is: suggestive evidence that the ETF adjusts toward the index, with wide bootstrap intervals (seminar B8)
- ▣ Information share of the index: between 0.25 and 0.98; residual correlation 0.93. Component share of the index 1.39, outside $[0, 1]$ because both α are negative
- ▣ Lesson: with daily closes the innovations are almost collinear, so the bounds are wide; price discovery needs intraday data (seminar B8)

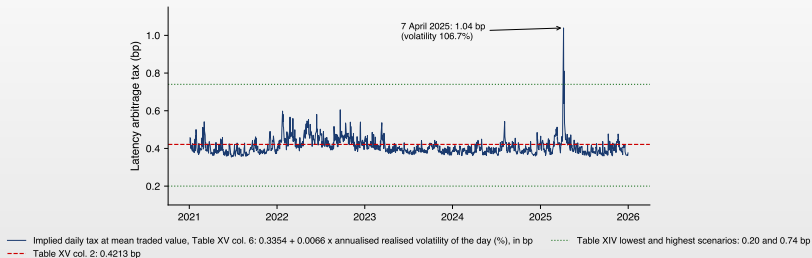
Case Study: the Speed Race and the Question

- **Latency arbitrage** (sniping): trading against quotes made stale by a public price move before their provider can cancel them [Budish, Cramton & Shim \(2015\)](#)
 - ▶ In a continuous book the fastest wins; liquidity providers pay through wider spreads and thinner books; proposal: **frequent batch auctions**, e.g. every second, moving competition from speed to price
- **Question** [Aquilina, Budish & O'Neill \(2022\)](#): how much does the race cost a whole stock market?
- Why it was hard: order-book data record only the winners; a failed order or a cancel that arrives too late leaves the book unchanged
 - ▶ **Message data**: every message to and from the exchange, including “too late” replies, so both winners and losers are seen
- Link to the section *The Bid-Ask Spread*: is sniping on **public** news a cost of liquidity next to **private**-information adverse selection (section *Information and Prices*)?

Case Study: Measurement and Three Central Results

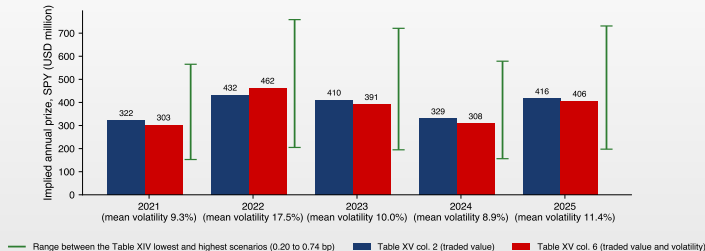
- **Measurement chain:** London Stock Exchange messages, FTSE 350, 43 days of 2015, about 2.2 billion messages (paper, Section II)
 - ▶ Race: 2+ participants, same stock, price and side, within the information horizon (at most 500 μ s), at least one success and one failure; race profit marked to the mid 10 s later (paper, Sections III, IV)
 - ▶ **Latency arbitrage tax:** daily race profits / daily traded value, in bp
- **1. Frequent and fast:** 537 races a day per FTSE 100 stock, about one a minute, decided by 5–10 μ s; 22% of FTSE 100 traded value is in races (paper, Tables II, III)
- **2. Concentrated:** the top six firms win 82% of races; a race is mostly a transfer from slower liquidity providers to a few fast takers (paper, Figures III, IV)
- **3. Small prizes, large sums:** tax 0.419 bp of all traded value (paper, Table X); eliminating races would cut the cost of liquidity by 16.73% (paper, Table XIII); USD 4.8 billion a year in global equities
- Complete definitions and results: [▶ Appendix](#)

Case Study: the Implied Daily Tax on SPY



- Tax $0.3354 + 0.0066 \sigma_t$ bp (paper, Table XV, col. 6) at mean traded value, with σ_t the annualised realised volatility (%) from SPY 5-minute bars, 1 244 full days of 2021–2025
- Median 0.40 bp, close to the 0.419 bp measured in London; peak 1.04 bp on 7 April 2025 ($\sigma_t = 106.7\%$); only 2 days above the highest scenario in the paper's Table XIV (0.74 bp)
- The volatility term matters only on stress days: most of the prize scales with traded value

Case Study: the Implied Annual Prize for SPY



- Annual prize $\sum_t \Pi_t$, USD million, Table XV col. 2 (0.4213 bp of traded value) and col. 6; whiskers: the 0.20–0.74 bp scenarios of Table XIV
- 2025: 416 (col. 2) and 406 (col. 6) million; col. 6 is larger only when the year's mean volatility exceeds $(0.4213 - 0.3354)/0.0066 = 13.0\%$, i.e. 2022 (17.5%)
- The scenarios differ by a factor of 3.7, the specifications by at most 7%: the tax level is the main uncertainty

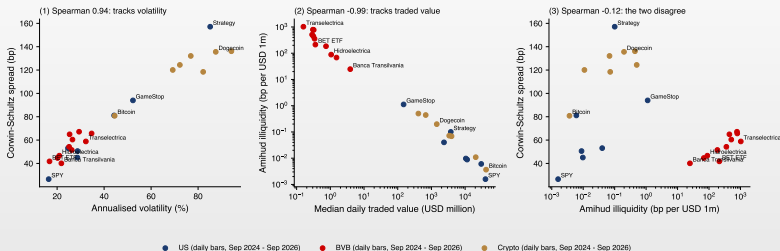
Case Study: What the Transfer Assumes

- **Method:** coefficient-transfer extrapolation: the London Table XV coefficients are applied to SPY's traded value and volatility; races are not re-estimated on SPY (no US message data)
 - ▶ It assumes the same race technology and the same tax per unit of value; volatility here is from 5-minute returns, the paper uses 1-minute returns
- **Limits stated by the authors:** one exchange, nine weeks; market structure (fragmentation, ETFs) and other asset classes ignored; long races and silent losers missed (paper, Sections V.A, VI.C)
 - ▶ For scale: the futures–SPY arbitrage of [Budish, Cramton & Shim \(2015\)](#) was about USD 75 million a year; the implied SPY prize is four to six times larger
- **Open extension** (seminar C3): does SPY's daily EDGE spread co-move with the implied tax? Both depend on volatility, so this is a diagnostic; identification needs a message-based race measure and a matched effective-spread benchmark

An Open Question: Do Price-Based Proxies Measure Trading Costs?

- Where trades and quotes are unavailable (BVB history, many crypto venues), costs are estimated from daily OHLC bars
 - ▶ Jahan-Parvar & Zikes (2023): low-frequency estimators track effective spreads only when the spread is large relative to volatility; otherwise volatility dominates
 - ▶ Ardia, Guidotti & Kroencke (2024): EDGE, an estimator from open, high, low and close prices, unbiased and efficient under their assumptions
- Formal question: $\hat{S}_i = S_i + b(\sigma_i) + \varepsilon_i$ across assets i
 - ▶ H_0 : $b \equiv 0$ (the proxy ranks assets by cost) against H_1 : b increasing in σ_i
 - ▶ Identification needs a cost benchmark measured independently of the proxy: effective spreads from matched trades and quotes; Amihud measures price impact, not the spread, and itself contains $|r|$
- Open: where direct cost data are scarce, is the spread-to-volatility ratio high enough for reliable estimation? Thin trading alone does not determine that ratio

Mini-Case: Cost or Volatility?



- A typical assistant claim: “the estimates are similar, so BVB blue chips are about as liquid as US large caps”; test on the 25 assets, Sep 2024 – Sep 2026 (Spearman)
- Corwin–Schultz with volatility $\rho = 0.94$ ($p < 0.001$), with Amihud $\rho = -0.12$ ($p = 0.58$); difference between the two Spearman correlations 1.06, 95% CI [0.58, 1.44] over assets, [0.71, 1.43] within markets; partial Corwin–Schultz–Amihud ρ controlling for volatility 0.27
- Daily Corwin–Schultz ranks track volatility, as Jahan-Parvar & Zikes (2023) predict; a diagnostic, not a test of H_0 : Amihud is not an independent spread benchmark



The Discovery Loop and a Project Seed

- **Loop**, with an AI assistant (prompts: ▶ Appendix)
 - ▶ 1. Literature: collect validation studies and check every DOI on [Crossref](#); 2. rival hypotheses, each with a cross-sectional prediction
 - ▶ 3. Replicate the chapter's numbers first (medians US 53, BVB 54, crypto 124 bp); 4. a pre-registered robustness plan; 5. ask the assistant to attack the benchmark; 6. report everything tried
- **Seed**: which estimator from daily bars, if any, measures the level of trading costs on the BVB and for crypto-assets?
- **Deliverables**
 - ▶ A dated pre-registration; EDGE implemented after [Ardia, Guidotti & Kroencke \(2024\)](#) and validated on simulated prices with a known spread; code and a two-page report with every specification tried
 - ▶ Declare AI use in `AI_USE.md` and the errors caught in `AI_ERRORS.md`: github.com/danpele/MFM/tree/main/project

Which Liquidity Measure When?

Available data	Suggested measure
Trades and quotes with timestamps	Effective and realised spread; MRR by GMM; Hasbrouck VAR; PIN
Prices of one asset on several venues	VECM; information-share bounds and component shares
Intraday bars (1–5 minutes)	Corwin–Schultz or Abdi–Ranaldo per day; intraday Amihud
Daily prices and volumes	Amihud ratio (price impact) across assets; EDGE for spread levels only if validated against effective spreads; high-low estimators mainly for changes over time
Own executions	Implementation shortfall; calibrate the square-root law
Theory, what-if analysis	Simulated order books; Glosten–Milgrom, Kyle, Almgren–Chriss

Check Yourself, and Next: Chapter 11

Retrieval questions

- Why can a positive Roll covariance occur even when the Roll model is true?
- Why do daily high-low estimators rank BVB stocks next to US large caps?
- In Kyle's model, what happens to λ and to the insider's profit when noise trading doubles?
- Why does a risk-averse seller front-load?

: continuous time

- From trades and order books back to continuous-time price models: Brownian motion and Itô calculus
- At high frequency the bid-ask bounce is microstructure noise that contaminates realised volatility ([Chapter 9](#))

Appendix: Order Types and the Simulated Book

- IOC, FOK, GTC: see the acronym list at the start of the chapter; **stop** order: becomes a market order once a trigger price trades; **iceberg**: displays only part of its size, the rest waits hidden
- A marketable limit order (a buy limit at or above the best ask) consumes liquidity like a market order, with a price cap
- **Zero-intelligence** simulation [Farmer, Patelli & Zovko \(2005\)](#)
 - ▶ Limit orders: rate α per price level, uniform up to 40 ticks from the opposite best quote; market orders: rate μ per side; cancellations: rate δ per standing order
 - ▶ Here $\alpha = 0.1$, $\mu = 0.5$, $\delta = 0.02$; 300 000 events, unit orders
 - ▶ Such a model already predicts much of the cross-section of spreads on the London Stock Exchange

Appendix: Statistical Properties of the Roll Estimator

- ▣ $\hat{\gamma}_1$: sample autocovariance of Δp from T price changes (demeaned); $\gamma_0 = \sigma^2 + 2c^2$, $\gamma_1 = -c^2$ Harris (1990)
 - Bias from demeaning: $\mathbb{E}[\hat{\gamma}_1] \approx \gamma_1 - (\gamma_0 + 2\gamma_1)/T$, negative even when $c = 0$
- ▣ Gaussian approximation (Bartlett's formula for an MA(1)): $T \text{Var}(\hat{\gamma}_1) \rightarrow \gamma_0^2 + 3\gamma_1^2$
- ▣ Hence, ignoring the demeaning bias, $P(\hat{\gamma}_1 > 0) \approx \Phi(\sqrt{T} \gamma_1 / \sqrt{\gamma_0^2 + 3\gamma_1^2}) \approx \Phi(\sqrt{T} \gamma_1 / \gamma_0)$ when $c^2 \ll \sigma^2$: close to 1/2 whenever $c^2 \ll \sigma^2 / \sqrt{T}$
- ▣ Delta method: $\text{se}(\hat{s}) \approx \sqrt{(\gamma_0^2 + 3\gamma_1^2)/T} / c$; $\hat{s} = 2\sqrt{-\hat{\gamma}_1}$ is also biased (Jensen's inequality, truncation at 0)
- ▣ With daily data the bounce (a few bp) is buried under the daily volatility (103 bp for SPY over the last two years), hence estimators based on the daily high and low
- ▣ Seminar A1–A2 derive these moments and the standard error step by step

Appendix: A Structural Model of Trading Costs: MRR

- Roll assumes i.i.d. trade signs; with first-order Markov signs, $\text{Corr}(q_t, q_{t-k}) = \rho^k$, independent of ε and without information, $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2(1 - \rho)^2$: Roll underestimates c
- MRR model [Madhavan, Richardson & Roomans \(1997\)](#): $p_t = m_t + \phi q_t + \xi_t$,
 $m_t = m_{t-1} + \theta (q_t - \mathbb{E}[q_t | q_{t-1}]) + \varepsilon_t$
 - ▶ θ : permanent (adverse-selection) impact of the order-flow **surprise**; ϕ : transitory cost (processing, inventory)
 - ▶ q_t a first-order Markov chain; public news ε_t and rounding noise ξ_t i.i.d., mean zero, independent of each other and of q
- With $u_t = \Delta p_t - (\phi + \theta)q_t + (\phi + \rho\theta)q_{t-1}$: GMM on $\mathbb{E}[q_t q_{t-1} - \rho q_{t-1}^2] = 0$, $\mathbb{E}[u_t - a] = 0$, $\mathbb{E}[(u_t - a)q_t] = 0$, $\mathbb{E}[(u_t - a)q_{t-1}] = 0$, a the drift (constant) of u_t
- Implied effective spread $2(\phi + \theta)$; adverse-selection share $\theta/(\phi + \theta)$
- Needs signed trades (trade-and-quote data); the public 5-minute bars do not identify q_t , so MRR is not estimated here

Appendix: Corwin–Schultz and EDGE in Practice

- Corwin–Schultz on daily data: the derivation assumes no price move overnight; the paper shifts H_{t+1} , L_{t+1} down by $L_{t+1} - C_t$ if $L_{t+1} > C_t$, up by $C_t - H_{t+1}$ if $H_{t+1} < C_t$ (C_t the close) [Corwin & Schultz \(2012\)](#)
- Within one session of 5-minute bars no overnight adjustment is needed; negative two-bar estimates are set to 0
- EDGE over windows: the authors recommend setting each negative window estimate to zero and then averaging, since more negative estimates go with larger spreads [Ardia, Guidotti & Kroencke \(2024\)](#)
- Truncation at zero biases the mean upwards: report the share truncated and a median

Appendix: Estimating PIN, and Why It Is Fragile

- EKOP model [Easley, Kiefer, O'Hara & Paperman \(1996\)](#): each day an information event occurs with probability α (bad news with probability δ); given the day's information state, daily buy and sell counts B and S are independent Poisson
 - ▶ Rates: uninformed $\varepsilon_b, \varepsilon_s$; informed μ added to the buy side on good-news days, to the sell side on bad-news days
- $L = (1 - \alpha) \pi(B; \varepsilon_b) \pi(S; \varepsilon_s) + \alpha \delta \pi(B; \varepsilon_b) \pi(S; \mu + \varepsilon_s) + \alpha(1 - \delta) \pi(B; \mu + \varepsilon_b) \pi(S; \varepsilon_s)$, π the Poisson pmf;

$$\text{PIN} = \frac{\alpha \mu}{\alpha \mu + \varepsilon_b + \varepsilon_s}$$
- Numerics: for liquid stocks ($B \approx 10^4$) $e^{-\varepsilon} \varepsilon^B / B!$ under- or overflows; factor out common terms and use log-sum-exp [Lin & Ke \(2011\)](#)
 - ▶ Without it, estimates pile up at boundary values; the likelihood is also multimodal: use many starting values (seminar A9–A10)

Appendix: The Hasbrouck Trade–Quote VAR

- ▣ VAR (vector autoregression) in the mid-quote return r_t and the signed trade x_t [Hasbrouck \(1991\)](#)
 - ▶ $r_t = \sum_{i \geq 1} a_i r_{t-i} + \sum_{i \geq 0} b_i x_{t-i} + \varepsilon_{1t}$, $x_t = \sum_{i \geq 1} c_i r_{t-i} + \sum_{i \geq 1} d_i x_{t-i} + \varepsilon_{2t}$
- ▣ Identification: the trade comes first within the interval (b_0 enters, the reverse contemporaneous term does not): a recursive (Cholesky) ordering
- ▣ **Permanent impact:** the long-run cumulative response of the quote to a trade shock, $\lim_{h \rightarrow \infty} \sum_{k=0}^h \partial r_{t+k} / \partial \varepsilon_{2t}$: the information content of trades
- ▣ Variance decomposition of the efficient price: the share explained by trades generalises the adverse-selection share of MRR
- ▣ Like MRR, it needs signed trades; in Kyle's model the permanent impact is exactly λ

Appendix: Order Flow Is Long-Memory and Self-Exciting

- Signs of market orders are autocorrelated with a power-law decay $\text{Corr}(\varepsilon_t, \varepsilon_{t+\ell}) \sim \ell^{-g}$, $0 < g < 1$: metaorders are split [Bouchaud, Gefen, Potters & Wyart \(2004\)](#)
 - ▶ Yet prices are close to a random walk: the propagator model $p_t = \sum_{s < t} G(t-s) \varepsilon_s + \text{noise}$ with a decaying kernel G reconciles the two
- Arrival times cluster: a Hawkes process has intensity $\lambda(t) = \mu + \int_{-\infty}^t \phi(t-s) dN_s$ [Bacry, Mastromatteo & Muzy \(2015\)](#)
 - ▶ Estimated by maximum likelihood; the branching ratio $\|\phi\|_1 = \int \phi$ is the share of events triggered by earlier events (endogeneity), stationary if < 1
- Links: durations between VaR exceedances and their clustering ([Chapter 8](#)), intraday volatility ([Chapter 9](#)), the concave impact (section *Price Impact and Optimal Execution*)

Appendix: Deriving the Almgren–Chriss Trajectory

- Continuous time, holdings $x(t)$, trading rate $-\dot{x}$: temporary impact cost $\eta \int_0^T \dot{x}^2 dt$, price-risk variance $\sigma^2 \int_0^T x^2 dt$
 - ▶ Minimise $\int_0^T (\eta \dot{x}^2 + \lambda \sigma^2 x^2) dt$ subject to $x(0) = X$, $x(T) = 0$ (the permanent impact $\frac{1}{2}\gamma X^2$ does not depend on the path)
- Euler–Lagrange for $F(x, \dot{x}) = \eta \dot{x}^2 + \lambda \sigma^2 x^2$: $\frac{d}{dt} \partial F / \partial \dot{x} = \partial F / \partial x \Rightarrow \ddot{x} = \kappa^2 x$, $\kappa^2 = \lambda \sigma^2 / \eta$
- General solution $Ae^{\kappa t} + Be^{-\kappa t}$; the boundary conditions give $x(t) = X \sinh(\kappa(T - t)) / \sinh(\kappa T)$
- In discrete time κ^2 becomes $\frac{2}{\tau^2} (\cosh \kappa \tau - 1) = \lambda \sigma^2 / \tilde{\eta}$ (section *Price Impact and Optimal Execution*); a resilient book with transient impact changes the shape [Obizhaeva & Wang \(2013\)](#)

Appendix: Almgren–Chriss, the Numerical Table

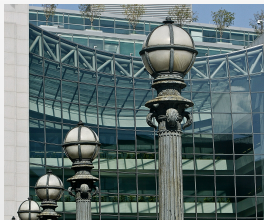
Strategy (5 daily steps)	Expected cost	Standard deviation
Even selling, $\lambda = 0$	0.60 million USD	1.04 million USD
$\lambda = 2 \cdot 10^{-6}$	1.08	0.45
$\lambda = 2 \cdot 10^{-5}$	2.05	0.10
$\lambda = 10^{-4}$	2.38	0.02

- With $\lambda = 2 \cdot 10^{-6}$, 428 599 shares remain after the first day
- In discrete time κ solves $\frac{2}{\tau^2}(\cosh \kappa \tau - 1) = \lambda \sigma^2 / \tilde{\eta}$; a resilient book with transient impact changes the shape [Obizhaeva & Wang \(2013\)](#)

Appendix: Optimal Market Making: Avellaneda–Stoikov

- A dealer with inventory q maximises expected exponential (CARA) utility of terminal wealth, risk aversion γ ; mid-price s_t is a Brownian motion with volatility σ [Avellaneda & Stoikov \(2008\)](#)
 - ▶ Market orders hit a quote at distance δ from the mid with intensity $Ae^{-k\delta}$
- Reservation (indifference) price: $r = s - q\gamma\sigma^2(T - t)$: a long dealer ($q > 0$) values the asset below the mid
- Approximate optimal total spread around r (from an expansion of the value function, not the exact solution):
$$\delta^a + \delta^b = \gamma\sigma^2(T - t) + \frac{2}{\gamma} \ln\left(1 + \frac{\gamma}{k}\right)$$
- Inventory skews both quotes: this formalises the inventory component of the spread (section *The Bid-Ask Spread*); the $\ln(1 + \gamma/k)$ term prices the thinness of the order flow

Appendix: The Flash Crash of 6 May 2010



Headquarters of the US Securities and Exchange
Commission, Washington

Photo: David (dbking) (2008); CC BY 2.0; Wikimedia
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- Report of the US futures (CFTC) and securities (SEC) regulators [CFTC & SEC \(2010\)](#)
- At 2:32 p.m. a mutual fund complex started selling 75 000 E-Mini S&P 500 futures (about USD 4.1 billion)
 - ▶ The algorithm targeted 9% of the previous minute's volume, "without regard to price or time"
- By 2:45:28 p.m. buy-side depth in the E-Mini had fallen below 1% of its morning level
- Over 20 000 trades in more than 300 securities were executed more than 60% away from their earlier prices, some at a penny
- Responses: single-stock circuit breakers (a 5-minute pause after a 10% move in 5 minutes) and clearer rules for cancelling erroneous trades [CFTC & SEC \(2010\)](#)

Appendix: The Bucharest Stock Exchange



Offices of the Bucharest Stock Exchange

Photo: [Za.kurek \(2014\)](#); CC BY-SA 4.0; [Wikimedia](#)

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- An electronic limit order book, as in the rest of this chapter [BVB \(history\)](#)
- Emerging-market status (FTSE Russell) since September 2020; Banca Transilvania and Nuclearelectrica were the first stocks in FTSE emerging-market indices
- A thin market: herding and bubbles in illiquid markets were studied on the BVB by [Pele, Mazurencu-Marinescu & Nijkamp \(2013\)](#)
 - ▶ Bubbles and crashes are the subject of [Chapter 17](#)

Appendix: Case Study: Data and Method

- **Data** (paper, Section II): London Stock Exchange message data for all FTSE 350 stocks, 17 August – 16 October 2015, 43 trading days, about 2.2 billion messages, obtained by the British regulator
 - ▶ Time stamps accurate to the microsecond (μs), taken where a message reaches the exchange's network, before any exchange processing
- **Race** (paper, Section III): 2+ participants on the same stock, price and side; 1+ take and 1+ cancel attempts, or 2+ takes; at least one success and one failure
 - ▶ "Same time": within the **information horizon**, the first message's observed processing time plus the fastest reaction time seen ($29 \mu s$), capped at $500 \mu s$
- **Race profit** (paper, Section IV.C): the signed gap between the race price and the mid-price 10 s later, times the displayed depth
- **Latency arbitrage tax** (paper, Section IV.D): daily race profits / daily traded value, in bp; if races vanished, the cost of liquidity would fall by the non-race tax / non-race effective half-spread (eq. 7)
- **Extrapolation** (paper, Section VI, Table XV): OLS of daily race profits on traded value V_t , and on V_t and 1-minute realised volatility σ_t times mean traded value $\bar{V}$, without a constant

Appendix: Case Study: Key Results

- **Frequent and fast** (paper, Table II, Figure II): 537 races a day for the average FTSE 100 stock, about one a minute; the winner beats the first loser by 5–10 μs (mode), 46 μs (median)
 - ▶ 22.15% of FTSE 100 traded value is in races (paper, Table III): speed decides a fifth of trading
- **Concentrated** (paper, Figures III, IV): the top six firms win 82% of races and take about 80% of the liquidity taken in races, but provide only 42% of it
 - ▶ A race is mostly a transfer from slower liquidity providers to a few fast takers
- **Small prizes, large sums:** per FTSE 100 race, 0.48 ticks (1.20 bp) per share and £1.95 in total (paper, Table VII); tax 0.419 bp of all traded value (paper, Table X); $R^2 = 0.811$ of profits on volume (paper, Table XV)
 - ▶ Global equities, 2018: USD 4.8 billion a year (range 2.3–8.4 across race definitions, Table XVII)
- **Cost of liquidity** (paper, Tables XI, XIV): in the full sample, race price impact is 30.58% of all price impact and 32.82% of the effective spread; FTSE 100 realised spread -1.93 bp in races, $+0.15$ bp outside (one-way spreads, half the Section 3 measures)
 - ▶ In the paper's model, eliminating races (e.g. frequent batch auctions) would cut the cost of liquidity by 16.73% (paper, Table XIII): sniping belongs next to adverse selection

Appendix: Case Study: Reading the Extrapolation

- **Daily tax:** median 0.40 bp, close to the 0.419 bp measured in London (paper, Table X); above col. 2's 0.4213 bp on 30% of days
 - ▶ Peak 1.04 bp on 7 April 2025 ($\sigma_t = 106.7\%$); only 2 days, 7 and 9 April 2025, lie above the 0.74 bp of the highest scenario
 - ▶ The volatility term moves the tax only on stress days: most of the prize scales with traded value
- **Annual prize:** USD 322–432 million a year under col. 2; col. 6 is larger only in 2022 (462 against 432)
 - ▶ Col. 6 exceeds col. 2 when the mean volatility of the year is above $(0.4213 - 0.3354)/0.0066 = 13.0\%$; 2022 averaged 17.5%, the other years 8.9–11.4%
 - ▶ The choice of specification moves the prize by at most 7%; the lowest and highest Table XIV scenarios differ by a factor of 3.7: the tax level, not the model, is the main uncertainty
- For scale: the single futures–SPY trade of [Budish, Cramton & Shim \(2015\)](#) was about USD 75 million a year; the implied SPY prize is four to six times larger

Appendix: The Discovery Loop, Reusable Prompts (1/2)

- Tools: an AI assistant (an LLM such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: [Semantic Scholar](#), [Elicit](#), [Google Scholar](#)
- 1. **Literature**: collect validation studies, then check every DOI on [Crossref](#) (the title must match)
 - ▶ Find studies that validate Roll, Corwin-Schultz, Abdi-Ranaldo and EDGE against effective spreads from trades and quotes; report the benchmark, the correlation in levels and in changes, the markets, and the DOI.
- 2. **Competing hypotheses**, each with a cross-sectional prediction
 - ▶ Give rival explanations for why Corwin-Schultz ranks BVB blue chips next to US large caps, and for each the predicted Spearman correlation of the estimate with volatility, with the Amihud ratio and with traded value.
- 3. **Data and code**: replicate the chapter's numbers first
 - ▶ Write Python that loads the daily OHLC prices and volumes of the 25 assets of this chapter and computes Corwin-Schultz, with the overnight adjustment and negative two-day estimates set to zero. Target medians: US 53, BVB 54, crypto 124 bp

Appendix: The Discovery Loop, Reusable Prompts (2/2)

- 4. **Robustness**, pre-registered
 - ▶ Write a pre-registered plan: Spearman and partial Spearman correlations, bootstrap over assets, leave-one-market-out (US, BVB, crypto), a second window 2022-2024, and EDGE and Abdi-Ranaldo in place of Corwin-Schultz.
- 5. **Critique**: ask the assistant to attack the benchmark itself
 - ▶ Act as a referee: argue that the Amihud ratio is not a cost measure either (it scales with volume and with volatility); which benchmark would identify the spread, and what would it cost to obtain?
- 6. **Report**: every estimator and window, bootstrap intervals, the benchmark's limits, code and the AI log

Appendix: What the Researcher Must Check

- **Tick-size benchmark:** no **quoted** spread is below one tick (effective spreads can be); compare every estimate with $0.01/P$ (0.21 bp for SPY) as a plausibility check, not a validation
- **Volatility contamination:** test: partial correlation with the benchmark controlling for volatility, and the 5-minute versus daily comparison in the section *The Bid-Ask Spread*
- **Truncation:** negative estimates set to zero bias the mean upwards; report the share truncated and a median
- **Units and venues:** lei converted at the BNR rate; crypto volume summed over exchanges, some with inflated volume; test: rerun on a subset of venues
- **Small, clustered cross-section:** 25 assets in 3 markets are not exchangeable, and 3 clusters are too few for cluster-robust inference; report the asset bootstrap as conditional on independence, plus a within-market bootstrap and leave-one-market-out
- **Mistyped formulas:** check the constants of the estimator against the paper and run a unit test on simulated prices with a known spread

Appendix: Further Study

Advanced courses on quantinar.com

-  Applied Computational Finance

Reading, code and data

- Textbooks: Hasbrouck (2007), Bouchaud, Bonart, Donier & Gould (2018), Harris (2003), Franke, Härdle & Hafner (2019)
- High-frequency Bitcoin data and daily risk forecasts: Pele & Mazurencu-Marinescu-Pele (2019)
- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_10

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